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Palaeomagnetism of some Cenozoic sediments, Cairo–Fayum area, Egypt

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Abstract

The palaeomagnetic and rock magnetic characteristics of some Cenozoic rocks from the Cairo–Fayum area have been investigated. A total number of 259 oriented core samples were collected at 32 sites located in rocks of Eocene (13 sites), Oligocene (11 sites) and Pliocene (9 sites) ages. Most of these rocks carry a weak but stable remanent magnetisation that is principally carried by hematite. Goethite and magnetite are also found in some samples as subordinate constituents. Careful thermal demagnetisation successfully enabled the isolation of the characteristic remanent magnetisation. Normal and reversed polarities that passed a reversal test have been recorded in the three age groups. This magnetisation is probably of primary origin and reflects the ages of the rocks. The resultant palaeomagnetic poles are considered reliable and represent a good contribution to the African palaeomagnetic database and should help in further refining of the Cenozoic APWP for Africa. © 1999 Elsevier Science B.V. All rights reserved.

Keywords: Paleomagnetism; Cenozoic sediments; Cairo–Fayum; Africa APWP

1. Introduction

Recent analyses of available African palaeomagnetic database (Besse and Courtillot, 1988, 1991; Scalera et al., 1996; Tarling and Abdeldayem, 1996) indicate that the Cenozoic Apparent Polar Wander Path (APWP) still lacks data in many parts. In some cases palaeomagnetic poles had, therefore, to be computed using a combination of poles and inclination data coming either from Deep Sea Drilling Projects (DSDP) cores or terranes that might have undergone tectonic rotations (Besse and Courtillot, 1991). It is, therefore, the aim of the present study to provide some reliable Cenozoic palaeomagnetic poles by studying the palaeomagnetic characteristics of some Tertiary sedimentary rocks that crop out in the Cairo–Fayum area. The rocks chosen for the present study are the type localities of the Eocene, Oligocene and Pliocene periods in Egypt. These are the Mokattam limestone (30.04°N, 31.26°E), the Qatrani Formation (29.6°N, 30.6°E) and the Kom El-Shellul Formation (29.9°N, 31.2°E), respectively.

The typical Mokattam Middle-Upper Eocene section lies to the east of Cairo facing the Citadel and is divided into two formations; the Mokattam Formation at the base and the Maadi Formation on top. The

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former comprises 170 m of numulitic limestone while the latter composes of 60 m of non-numulitic or sparsely numulitic sandy limestone and sandy dolomitic limestone (Strougo, 1985; Zalat, 1987). The area is structurally simple with few high angle normal faults that dissect the brittle limestone of the Mokattam Formation while only gently drap the ductile limestone of the overlying Maadi Formation forming monoclines with dips $\leq 3°$ (Moustafa and Abdel Tawab, 1985; Moustafa et al., 1985). The Oligocene Qatrani Formation is composed of c. 350 m of nearly horizontal fluviomarine complex of variegated rocks comprising fine to coarse sandstone, granule and pebbly conglomerate, sandy mudstone and limestone (Bown and Kraus, 1988). This formation is known world-wide for its rich and diverse mammalian fauna (e.g., Simons, 1990) and has long been known to be of Oligocene age (Blanckenhorn, 1903; Said, 1962). Some recent studies (e.g., Van Couvering and Harris, 1991; Rasmussen et al., 1992; Kappelman et al., 1992), however, argued for a Late Eocene to Early Oligocene age to sediments of this formation. Most recently, Gingerich (1993) and Domning et al. (1994) reinforced an Oligocene age based on a detailed study of sea level sequence stratigraphy and the presence of a new Mammalia species. The Widan el Faras basalt that unconformably caps this formation has been radiometrically dated several times using conventional whole rock methods yielding ages ranging between 25.3 to 31.0 Ma (e.g., Simons and Wood, 1968; Fleagle et al., 1986). More recently, Kappelman et al. (1992) assigned a new date of 23.64 ± 0.035 Ma using the laser incremental heating $^{40}Ar/^{39}Ar$ method. Finally, the early Pliocene Kom El-Shellul Formation crops out in the structurally simple area to the NW of the Sakkara Pyramid (Giza). Its section consists of about 13 m of horizontally bedded sandy detrital limestone that was deposited in agitated shallow to littoral sea and arid climate (Youssef et al., 1984).

2. Methodology

Standard 1 in. diameter core samples were collected in the field using a portable gasoline-powered rock drill and were individually in situ oriented using either a sun compass or a magnetic compass. All samples were then sliced to the optimum 2.2 cm length.

In the laboratory, NRM was measured using a Minspin MS2 and a Jelinek JR4 spinner magnetometers. Alternating field (A.F.) and thermal demagnetisations were carried out using a Molspin shielded demagnetiser with two-axis tumbler and a peak field of 100 mT and a shielded Schonstedt type furnace capable of heating up to 25 samples, respectively. A Molspin pulse magnetiser producing unidirectional fields up to 800 mT was used for isothermal remanent magnetisation (IRM) experiments.

Both visual (using stereographic and orthogonal plots) and statistical (using the principal component analysis of Kirschvink, 1980) methods were used for analysis of demagnetisation data. Mean directions were computed using Fisher's (1953) statistics.

The magnetic mineralogy was investigated by studying the coercivity IRM acquisition spectrum and the unblocking temperature characteristics of a laboratory imposed axial IRM.

3. Sampling

A total number of 259 oriented core samples were collected at 32 sites from the studied three sections. Thirteen sites (comprising 115 samples) were drilled in the Mokattam Eocene limestone (eight sites from the Mokattam Formation and five sites from the Maadi Formation). Nine sites (comprising 70 samples) were collected from the Qatrani Formation. It is worth noting that sampling was not systematically distributed throughout this formation as some parts were either too coarse (mainly in the lower part of the section) or too soft (mainly in the middle part) for drilling. Two extra sites (17 samples) were sampled from the Widan el Faras basalt flows capping the Qatrani Formation. The remaining eight sites (57 samples) were from the Kom El-Shellul Pliocene limestone. Samples were taken from fresh and undisturbed exposures.

4. Magnetic mineralogy

Two groups of representative samples covering different lithologic varieties from the three studied

sections were chosen for studying the magnetic minerals responsible for carrying the magnetisation. The first group of samples were given stepwisely increasing direct fields from 10 mT to a maximum of 800 mT to determine their IRM acquisition pattern. The majority of samples from the three sections showed a similar behaviour characterised by a slow rate of IRM acquisition with no saturation up to the maximum field (EO 1.7, OL 5.3 and PL 6.7, Fig. 1a). Some other samples (EO 8.2 and PL 1.3, Fig. 1a) showed an initial rapid acquisition of about half of the total IRM at fields ≤ 150 mT followed by a continuous slow rate with no saturation up to the end of experiment. The basalt samples together with few

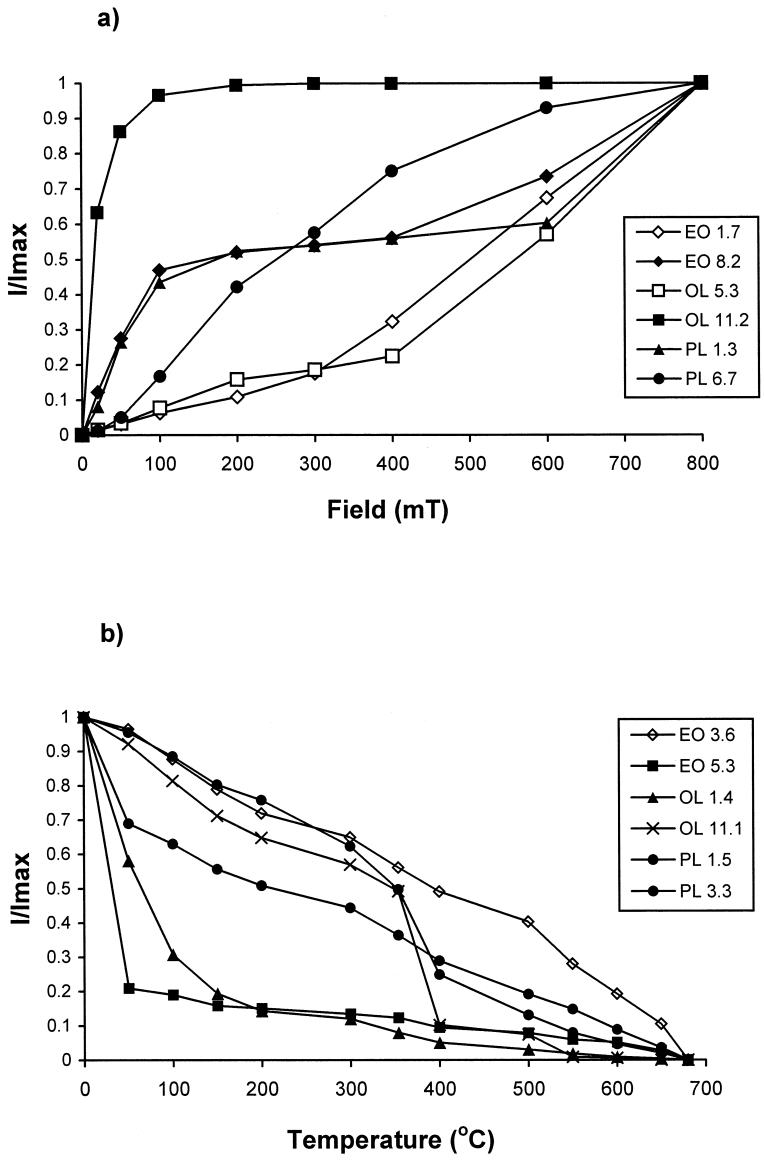


Fig. 1. Normalised curves for (a) IRM acquisition and (b) thermal demagnetization of axial IRM for representative samples from the three studied sections.

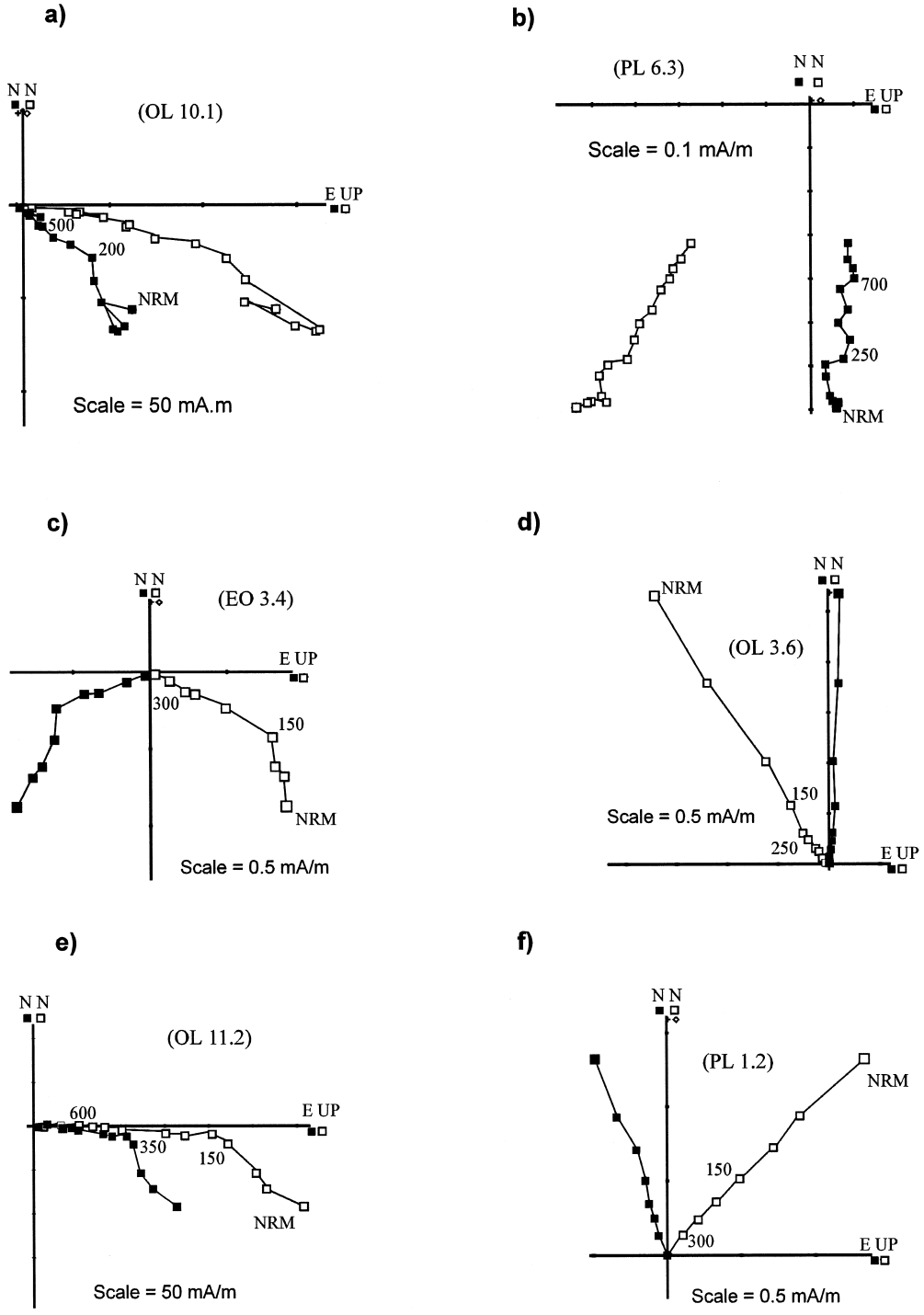


Fig. 2. Typical examples of orthogonal plots (Zijderveld, 1967) of progressive A.F. demagnetisation data of representative samples from (a) Widan el Faras basalt, (b) Sakkara Pliocene limestone and stepwise thermal demagnetisation data for representative samples from (c) Mokattam Eocene limestone, (d) Qatrani Oligocene complex, (e) Widan el Faras basalt, and (f) Sakkara Pliocene limestone. Closed (open) symbols are projections on horizontal (vertical) planes.

samples from the Eocene rocks, on the other hand, showed a distinct behaviour characterised by an initial rapid rate of acquisition of about 90% of the total IRM at fields ≤ 100 mT followed by a complete saturation at higher fields (OL 11.2, Fig. 1a).

Thermal demagnetisation of a laboratory imposed axial IRM given to the other group of selected samples gave more details. The larger group of samples from the three sections showed similar behaviour with their remanence steadily decaying up to 680°C (EO 3.6 and PL 1.5, Fig. 1b). Another group of samples from the Eocene and Oligocene sections displayed an initial loss of 50–80% of their total IRM at temperature ≤ 150°, then slowly continued to loose the remaining magnetisation until it was completely destroyed by 680°C (EO 5.3 and OL 1.4, Fig. 1b). The basalt samples and some other few samples from the Oligocene and Pliocene sections showed a different behaviour characterised by a uniform decay of their remanence up 350°C when the intensity dropped then continued to decay until completely destroyed by 580°C for the basalt samples and 680°C for the sedimentary samples (OL 11.1 and PL 3.3, Fig. 1b).

The combined results from both IRM acquisition and thermal demagnetisation indicate that although hematite is the principal magnetic carrier in most samples, goethite, titanomagnetite and magnetite are also present as subordinate constituents.

5. Palaeomagnetic results and analysis

Initial NRM measurements gave low to very low intensities for the Mokattam Eocene rocks; averaging 0.08 mA/m for samples from the lower part of the Mokattam Formation and all those from the Maadi Formation and 0.53 mA/m for samples from the middle and upper parts of the Mokattam Formation. Except for the basalt samples with an average intensity of 270.8 mA/m, the majority of the Qatrani Oligocene samples had low intensities averaging 1.8 mA/m. Similarly, the Sakkara Pliocene samples yielded very low (average of 0.12 mA/m) to low (average of 0.83 mA/m) intensities from the lower and upper parts of the section, respectively.

Table 1
Site mean characteristic NRM directions and VGPs for the Eocene, Oligocene and Pliocene rocks

Site	N	Dec. (°)	Inc. (°)	k	α_{95} (°)	VGP	
						Lat. (°)	Long. (°)
Sakkara Pliocene limestone							
1	6	177	−41	62	7.8	−83	53
2	6	8	46	75	7.1	83	138
3	7	11	42	51	8.5	78	149
4	7	19	39	46	9.0	71	141
5	6	191	−48	47	8.9	−80	302
6	6	23	42	131	5.7	69	132
7	6	182	−49	103	5.8	−89	293
8	7	358	46	58	6.8	87	245
Mean	8	9	45	112	5.3	82	144
Qatrani Oligocene complex							
1	7	177	−41	46	8.9	−84	54
2	8	22	46	90	5.9	71	121
3	7	358	36	37	10.1	80	224
4	6	200	−39	75	7.8	−71	320
5	6	10	43	52	8.5	80	144
6	8	17	39	45	9.2	73	143
7	7	206	−46	40	9.0	−67	300
8	7	178	−37	231	6.0	−81	41
9	8	3	41	121	4.9	84	184
10[b]	8	201	−56	48	8.3	−71	273
11[b]	7	212	−54	25	9.8	−62	282
Mean	9	10	42	77	5.9	80	151
Mean[b]	2	207	−55	271	15.3	−67	278
Mokattam Eocene limestone							
1	9	16	39	126	5.6	73	146
2	7	359	42	38	9.2	84	214
3	8	179	−36	101	5.2	−80	36
4	8	190	−34	73	8.1	−75	349
5	8	18	42	121	5.1	73	137
6	9	8	36	175	4.0	78	173
7	9	22	39	78	6.7	69	139
8	8	181	−35	90	6.0	−79	28
9	8	4	42	31	9.4	83	183
10∗	8	62	56	22	19.1	39	97
11	9	10	34	38	9.2	76	172
12	8	200	−43	47	8.1	−71	313
13∗	7	32	59	33	16.2	62	91
Mean	11	10	39	123	4.1	78	163

N = number of samples in each site (or sites for mean calculation).
Dec. and Inc. = magnetic declination and inclination.
Lat. and Long. = latitude and longitude of VGP.
k and α_{95} = precision parameter and semi-angle of confidence (Fisher, 1953).
∗Site excluded from overall mean calculations.
[b]Sites from the Widan el Faras basalt and their mean.
Figures are rounded to the nearest degree.

The stability of NRM was tested against both A.F. and thermal demagnetisations by first subjecting two representative samples from each site to a full range of treatment for each of two demagnetisation methods. A.F. was first applied at incremental steps up to 100 mT. The majority of samples from the Eocene and Oligocene sections showed a very high stability against A.F. with both intensities and directions erratically fluctuating around their initial values. Only the representative Widan el Faras basalt sample responded significantly to A.F. with intensity decreasing continuously and directions, apart from the first few steps, linearly heading toward the origin of the orthogonal plot (Fig. 2a). The representative Pliocene samples responded partially to A.F. with intensities dropping to values $< 50\%$ of their initial values and directions tend to display a relatively linear components that generally head toward the origin (Fig. 2b).

Unlike A.F., thermal demagnetisation at incremental steps up to 680°C had a significant effect on samples from the three sections. Physico-chemical changes associated commonly with heating had been monitored through measurements of magnetic susceptibility values at each step. Most of the Eocene samples showed a continuous steady decay of their remanence up to ≤ 450°C when it was destroyed resulting, in most cases, in the separation of two components; a low temperature component ($<$ 200°C) and a high temperature linear component (Fig. 2c). Some of the Oligocene samples behaved similar to the Eocene samples while the others displayed an initial drop of intensity to values $< 30\%$ of the initial values at temperatures < 150°C then began to linearly loose the remaining magnetisation displaying a linear component that heads toward the origin (Fig. 2d). The representative basalt sample gave rather similar results to that obtained during A.F. demagnetisation but with much better defined linear component that reaches the origin of the orthogonal plot (Fig. 2e). Finally, the majority of the Pliocene samples displayed a continuous steady decay of the remanence up to ≤ 300°C when it was destroyed; resulting, in most cases, in a single component that heads toward the origin (Fig. 2f).

In light of the visual and statistical analyses of the A.F. and thermal demagnetisation data and the results of the study of Walton (1996), it was decided

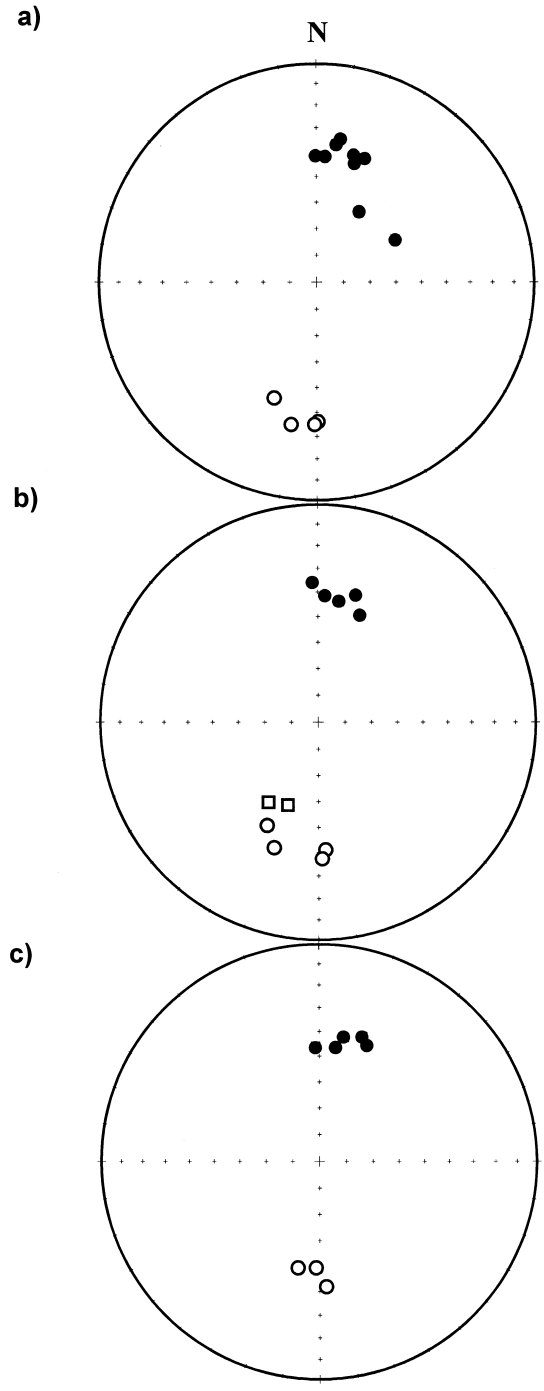


Fig. 3. Equal-area stereographic projections of site-mean characteristic remanent magnetisation of (a) the Mokattam Eocene limestone, (b) the Qatrani Oligocene complex (open squares are sites from the Widan el Faras basalt), and (c) the Sakkara Pliocene limestone.

that all remaining samples from the three studied groups should be treated thermally as linear component were much better defined. Consequently, all remaining samples from the three sections were subjected to a full range of thermal demagnetisation. Similar to results obtained during thermal demagnetisation of representative samples, many samples from the studied three sections yielded two components; a low temperature component ($\leq 200°C$) and a high temperature component. In most cases, the isolated low temperature component was mostly close to the direction of the present Earth's magnetic field at each of the studied areas and was therefore considered as a recently developed viscous remanence. The computed within- and between-samples principal component data (Kirschvink, 1980) for the high temperature component, characteristic magnetisation, were then compared and best estimated site mean directions were then computed (Fisher, 1953; Table 1). All three sections yielded both normal and reversed polarities (Table 1, Fig. 3). Two sites from the Mokattam section (10 and 13) were clearly deviant from the remaining sites that showed a relatively good cluster (Fig. 3a). These two sites were

Table 2
Selected Cenozoic north palaeomagnetic poles for Africa

Number	Rock unit, region	Age	Pole (Lat/Long)	Dp/Dm	Reference
1	Haruj Assuad Volcanics, Libya (27.8N/17.3E)	1	83/171	5/8.5	Ade-Hall et al., 1974
2	Jos Plateau Newer Basalts, Nigeria (9.6N/8.8E)	1	67/242	2.4/4.8	Garde and Ayeni, 1991
3	Volcanics of Afars and Issas 1, Djibouti (12N/43E)	1	82/253	5/10	Pouchan and Roche, 1971
4	East African Volcanics, Kenya and Tanzania (0.0/36E)	1	89/104	3.2/3.2	Reilly et al., 1976
5	Quaternary lavas, Morocco (32.5N/355E)	1	77/96	8/8	Najid et al., 1981
6	Ngorongoro Caldera, Tanzania (3.2S/35.5E)	2.5	81/62	6/12	Gromme et al., 1971
7	East African Volcanics, Kenya and Tanzania (0.0/36E)	3.5	87/148	2.3/2.3	Reilly et al., 1976
8	Sakkara sediments, Egypt (29.9N/31.2E)	4	82/144	4.2/6.7	This study
9	Garian Volcanics, Libya (32N/13E)	4	88/123	3.2/6.7	Ade-Hall et al., 1975
10	Volcanics, Danakil Block, Ethiopia (12.7N/42.5E)	4	80/258	2.6/2.6	Schult, 1974
11	Ait Kandoula Basin Sediments, Morocco (31N/353E)	8.5	86/286	4.8/7	Benammi et al., 1996
12	Miocene Volcanics, Morocco (35N/357E)	9	86/162	6.4/6.4	Najid et al., 1981
13	Ngorora Formation, Kenya (1N/35.5E)	11.5	86/256	1.9/3.8	Deino et al., 1990
14	Volcanics, Jebel Soda, Libya (28.7N/15.6E)	11.5	78/196	7.4/7.4	Schult and Soffel, 1973
15	East African Volcanics, Kenya and Tanzania (0.0/36E)	12	87/187	6.1/6.1	Reilly et al., 1976
16	Volcanics, Kenya (1.6S/35.9E)	13.5	80/34	8.9/8.9	Patel and Raja, 1979
17	Trap Series, Ethiopia (14N/39E)	14	73/254	2.4/4.8	Burek, 1974
18	Qattara sediments, Egypt (30.3N/28.8E)	17	77/198	1.4/2.6	Abdeldayem, 1996
19	Turkana Lavas, Kenya (0.0/36E)	18.5	85/163	2.3/2.3	Reilly et al., 1976
20	Massif de Cavallo, Algeria (32N/5E)	19	87/23	2.2/3.3	Bobier and Robin, 1969
21	Cairo Region Basalts, Egypt (30N/31E)	19.5	66/167	2.3/2.3	Lotfy et al., 1995
22	Cairo Region Basalts, Egypt (30N/31E)	19.5	76/111	3/3	Lotfy et al., 1995
23[a]	Qatrani Basalts, Egypt (29.6N/30.6E)	23	67/98	15.5/21.8	This study
24	Qatrani Basalt, Egypt (29.8N/30.7E)	26	73/81	12.7/12.7	Schult et al., 1981
25	Basalts, Abu Zaabal and Qatrani, Egypt (30.2N/31.2E)	26	64/80	8.9/11.2	El Shazly and Krs, 1971
26	Qatrani sediments, Egypt (29.6N/30.6E)	29	80/151	4.4/7.2	This study
27	Southern Plateau Volcanics, Ethiopia (9.1N/41E)	34	75/170	6.4/6.4	Schult, 1974
28	Baharia Iron ores, combined, Egypt (28.2N/28.9E)	37	84/139	7/7	Schult et al., 1981
29	Mokattam sediments, Egypt (30N/31.3E)	42	78/163	2.9/4.9	This study
30	Volcanics, Jebel Nefousa, Libya (32N/13.4E)	43	86/152	3.7/3.7	Schult and Soffel, 1973
31	Basalts, Wadi Abu Tereifiya, Egypt (30N/32.1E)	44.5	69/189	3.2/6.1	Hussain et al., 1979

Age is the mean radiometric/stratigraphic age. Lat/Long are palaeopole latitude and longitude.
Dp and Dm are the half angle of confidence on the palaeopole.
Pole positions are rounded to the nearest degree.
[a] Pole excluded from age window mean pole calculations (see text for details).

excluded from further calculations. The Qatrani sites showed a good cluster with the two basalt sites having steeper negative inclinations than the rest of sites (Fig. 3b) and were, therefore, treated separately on the basis that secular variation might not have been averaged out. The Pliocene sites similarly displayed a good cluster (Fig. 3c).

In order to test whether the obtained normal and reversed site mean directions in each of the three sections are antiparallel and drawn from the same

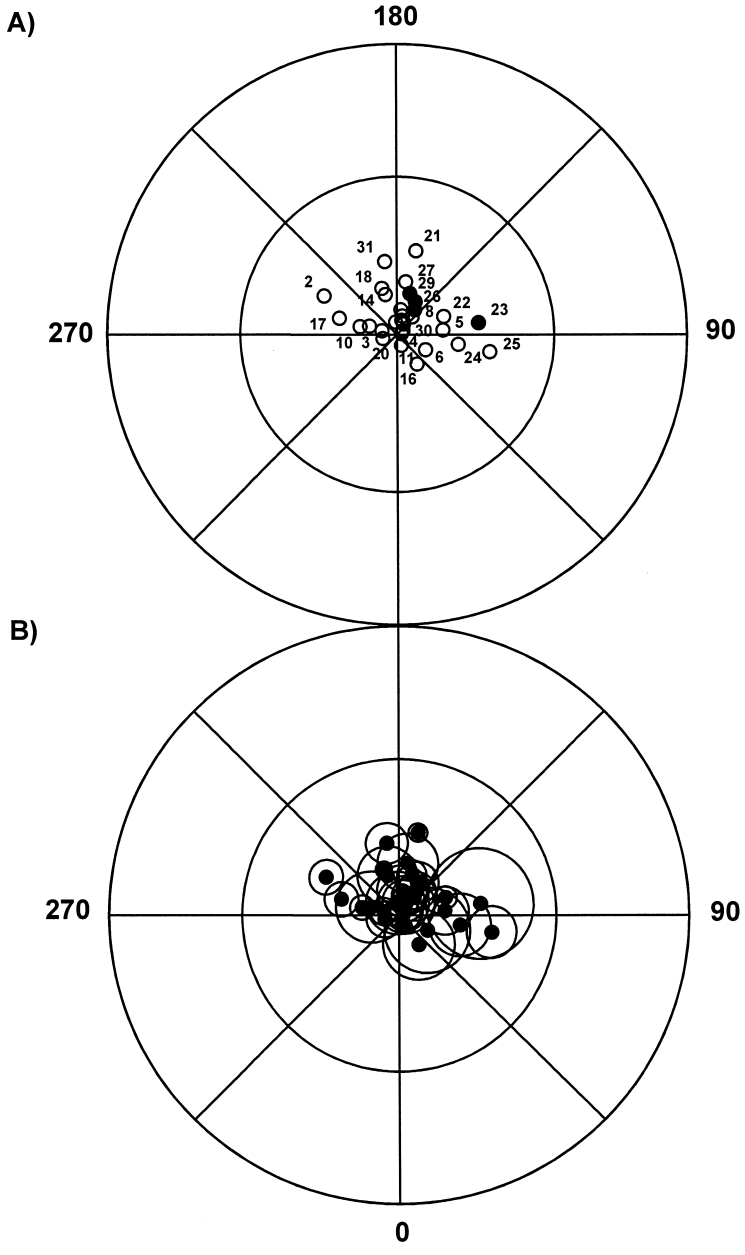


Fig. 4. (a) Previously published Cenozoic north paleomagnetic poles from Africa together with poles from the present study (closed circles). Numbers are according to the list in Table 2, and (b) same poles with their cones of confidence.

population, a reversal test (McFadden and McElhinny, 1990) was applied to each of three groups. The angle between the means of the two polarity sets was 3.7° with a critical angle of 10.2° for the Mokattam sites (excluding sites 10 and 13), 0.2° with a critical angle of 14.0° for the Qatrani sites (excluding the two basalt sites) and 7.1° with a critical angle of 10.2° for the Sakkara sites. This indicates that, under the null hypothesis, directions from each of the three groups seem to have been drawn from populations with means that are 180° apart and their directions could be resolved to better than 20° but not better than 10°. This classification is, therefore, positive in all cases and is of 'C' category (McFadden and McElhinny, 1990).

6. Discussion and implications for Africa's Cenozoic APWP

The present results reveal that the Mokattam Eocene limestone, Qatrani Oligocene sandstone and Sakkara Pliocene limestone pertain a weak but stable remanent magnetisation. Hematite is the principal magnetic carrier in most parts of the three studied sections while goethite and magnetite are subordinate magnetic constituents. The presence of goethite in some of the Eocene and Oligocene samples implies that some parts of these two sections have been subjected to extensive physico-chemical weathering.

Stepwise annealing of the entire collection has successfully helped in the separation of two components in many samples; a low temperature component (≤ 200°C) and a high temperature one. The former is mostly aligned with the present Earth's magnetic field and has probably been acquired due to recent chemical weathering, manifested by the presence of goethite in some samples. The latter is found to be stable and deviant from the present Earth's magnetic field. The presence of normal and reversed polarities pertaining to same populations in all cases suggests that the characteristic remanent magnetisation isolated from the three age groups is older than 700,000 years and is not associated with axial dipole swings or short lived dipole excursions recently recognised in some Tertiary rocks (Westphal, 1993). This magnetisation could therefore be considered of primary origin and reflects the ages of the rocks under consideration. The computed palaeomagnetic pole from the two basalt sites although not statistically significant it could be considered as a preliminary contribution for a future detailed study to be carried out. The resultant palaeomagnetic poles from the present study are considered reliable and represent a good contribution to the Cenozoic palaeomagnetic database of Africa.

Attempts of constructing a Cenozoic APWP for Africa have always been hampered by the lack of reliable data in many parts. In the present study the published Cenozoic palaeomagnetic poles for Africa

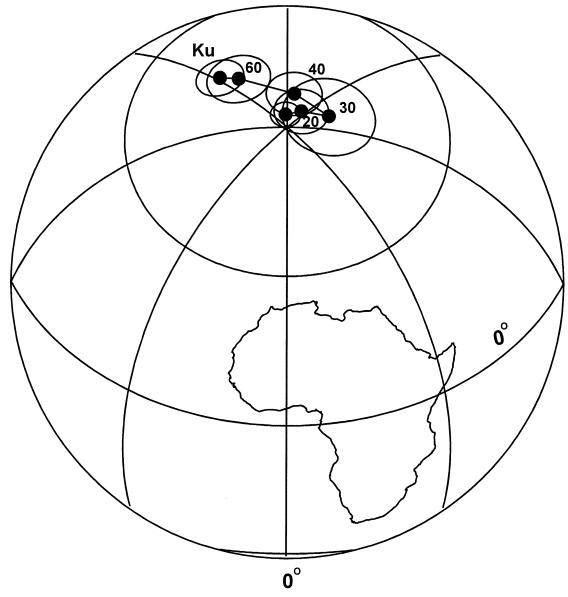


Fig. 5. Cenozoic apparent polar wander path for Africa constructed from mean paleopoles in Table 3. Ku is Upper Cretaceous paleopole for Africa (Besse and Courtillot, 1991; Table 1b) plotted for reference only.

Table 3
African Cenozoic mean poles derived from data in Table 2

Age Window	Ma	Plat	Plong	*N*	A_{95}
0–20	9	86	187	22	4.0
10–30	18	84	145	13	7.2
20–40	30	78	112	5	12.4
30–50	40	79	171	5	7.2
50–70	62	69	216	5	8.0

Ma is the mean age computed.
Plat and Plong are pole latitude and longitude, respectively.
N is the number of poles used.
The 62 Ma pole is that of Besse and Courtillot, 1991, Table 1b.
Numbers are rounded to the nearest degree.

have been reviewed. They were extracted from the most recently released version of the GPMDB, version 3.2 (McElhinny and Lock, 1996). Data from remagnetised and/or tectonically disturbed studies were eliminated. Only poles that have a Q value ≥ 3 (Van der Voo, 1990) with $A_{95} < 15^{\circ}$ and age that does not exceed a geologic period are considered. These poles together with those from the present study are listed in Table 2 and plotted in Fig. 4. The high scatter of the poles and overlap of their circles of confidence are clearly observed (Fig. 4b). This might be attributed to the inherited problem of incomplete averaging-out of secular variation as most of these poles have come from volcanic rocks (Table 2). On contrary, the three poles from the present study (solid symbols in Fig. 4a) seem to be consistent in defining a short APWP segment during the Tertiary. The pole from the two basalt sites, on the other hand, plots close to equivalent poles derived from volcanic rocks from Egypt and other parts in Africa (Table 2, Fig. 4a). This pole was eliminated from the mean pole calculations discussed below.

Bearing in mind the difficulty associated with the present data set, an attempt has been made to construct a Cenozoic APWP for Africa. Unlike some previous attempts (e.g., Besse and Courtillot, 1991; Tauxe et al., 1983), only data from the stable African craton (Table 2) have been used. Mean poles for appropriate time windows were computed (Table 3) and plotted together with their circles of confidence (Fig. 5). Yet no reliable data are available for the early Cenozoic and the 60 Ma mean pole of Besse and Courtillot (1991) (Table 1b) had to be used. It is immediately clear that although an APWP can be traced out there is still remaining some degree of uncertainty of its credibility. This is due to the convergence of relatively large circles of confidence around the poles, particularly that of the 30 Ma pole. The present attempt clearly demonstrates the desperate need for more reliable palaeomagnetic data, particularly from well-dated sediments where secular variation can be averaged out.

7. Conclusions

The following conclusions can be drawn from the present study.

(1) The Mokattam Eocene limestone, Qatrani Oligocene sandstone and Sakkara Pliocene limestone carry a weak but stable remanent magnetisation.

(2) Hematite is the principal magnetic carrier of magnetisation in all of three studied sections while goethite and magnetite are subordinates. Magnetite and titanomagnetite are the main carriers present in the Qatrani basalt flows.

(3) Careful thermal demagnetisation has made it possible to isolate the characteristic remanent magnetisation. This magnetisation is probably of primary origin and the resultant palaeomagnetic poles are considered reliable and represent a good contribution to the African Cenozoic palaeomagnetic database.

(4) Revision of the current African Cenozoic palaeomagnetic database demonstrates the necessity of more reliable palaeomagnetic data from well-dated sediments for further refinement of the African Cenozoic APWP.

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References

Abdeldayem, A.L., 1996. Palaeomagnetism of some Miocene rocks, Qattara depression, Western Desert, Egypt. J. Afr. Earth Sci. 22, 525–533.

Ade-Hall, J.M., Gerstein, S., Gerstein, R., Reynolds, P., Dagley, P., Mussett, A., Hubbard, T., 1975. Geophysical studies of North African Cenozoic volcanic areas: III. Garian, Libya. Can. J. Earth Sci. 12, 1264–1271.

Ade-Hall, J.M., Reynolds, P.H., Dagley, P., Mussett, A.E., Hubbard, T.P., Klitzsch, F., 1974. Geophysical studies of North African Cenozoic volcanic areas: I. Haruj Assuad. Can. J. Earth Sci. 11, 998–1006.

Benammi, M., Calvo, M., Prévot, M., Jaeger, J.J., 1996. Magne-

tostratigraphy and paleontology of Ait Kandoula Basin (High Atlas, Morocco) and the African–European late Miocene terrestrial fauna exchanges. Earth Planet. Sci. Lett. 145, 15–29.

Besse, J., Courtillot, V., 1988. Paleogeographic maps of the continents bordering the Indian Ocean since the Early Jurassic. J. Geophys. Res. 93, 11791–11808.

Besse, J., Courtillot, V., 1991. Revised and synthetic apparent polar wander paths of the African, Eurasian, North American and Indian plates, and true polar wander since 200 Ma. J. Geophys. Res. 96, 4029–4050.

Blanckenhorn, M., 1903. Neue geologisch-stratigraphische Beobachtungen in Aegypten. Sitzungsberichte der Mathematisch-physikalischen Classe der Königlichen bayerischen Akademie der Wissenschaften, München 32, 353–433.

Bobier, C., Robin, C., 1969. Etude paleomagnetique du Massif eruptif de Cavallo (Nord-Constantinios, Algerie). C. R. Acad. Sci. Paris Ser. D 269, 134–137.

Burek, P.J., 1974. Plattentektonische probleme in der weiteren umgebung Arabiens sowie der Danakil-Afar-Senke. Geotekt. Forsch 47, 1–93.

Bown, T.M., Kraus, M.J., 1988. Geology and paleoenvironment of the Oligocene Jebel Qatrani Formation and Adjacent rocks, Fayum Depression, Egypt. US Geol. Survey Prof. Paper 1452, 60 pp.

Deino, A., Tauxe, L., Monaghan, M., Drake, R., 1990. 40Ar/39Ar age calibration of the litho- and paleomagnetic stratigraphies of the Ngorora Formation, Kenya. J. Geol. 98, 567–587.

Domning, D.P., Gingerich, P.D., Simons, E.L., Ankel-Simons, F.A., 1994. A new Early Oligocene Dugongid (Mammalia, Sirenia) from Fayum Province, Egypt. Contib. Museum Plaeont. Univ. Michigan 29, 89–108.

El Shazly, E.M., Krs, M., 1971. Magnetism and palaeomagnetism of Oligocene basalts from Abu Zaabal and Qatrani, northern Egypt. Geofys. Sbornik 19, 261–270.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. R. Soc. London A 27, 295–305.

Fleagle, J.G., Bown, T.M., Obradovich, J.D., Simons, E.L., 1986. How old are the Fayum primates? In: Else, J.G., Lee, P.C. (Eds.), Primate Evolution. Cambridge Univ. Press, pp. 4–17.

Garde, S.C., Ayeni, I.O., 1991. A palaeomagnetic study of Cainozoic basalts from northern Nigeria. Geophys. J. Int. 106, 717–723.

Gingerich, P.D., 1993. Oligocene age of the Gebel Qatrani Formation, Fayum, Egypt. J. Hum. Evol. 24, 207–218.

Gromme, C.S., Reilly, T.A., Mussett, A.E., Hay, R.L., 1971. Palaeomagnetism and Potassium–Argon ages of Volcanic rocks of the Ngorongoro Caldera, Tanzania. Geophys. J. R. Astr. Soc. 22, 101–115.

Hussain, A.G., Schult, A., Soffel, H., 1979. Palaeomagnetism of the basalts of Wadi Abu Tereifiya, Mandisha and dioritic dykes of Wadi Abu Shihat, Egypt. Geophys. J. R. Astr. Soc. 56, 55–61.

Kappelman, J., Simons, E.L., Swisher III, C.C., 1992. New age determination for the Eocene–Oligocene boundary sediments in the Fayum depression, northern Egypt. J. Geol. 100, 647–668.

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of palaeomagnetic data. Geophys. J. R. Astr. Soc. 62, 699–718.

Lotfy, H.I., Van der Voo, R., Hall, C.M., Kamel, O.A., Abdel Aal, A.Y., 1995. Paleomagnetism of Early Miocene basalt eruptions in the areas east and west of Cairo. J. Afr. Earth Sci. 21, 407–419.

McElhinny, M.W., Lock, J., 1996. IAGA paleomagnetic databases with Access. Surv. Geophys. 17, 575–591.

McFadden, P.L., McElhinny, M.W., 1990. Classification of the reversal test in palaeomagnetism. Geophys. J. Int. 103, 725–729.

Moustafa, A.R., Abdel Tawab, S., 1985. Morphostructures and non-tectonic structures of Gebel Mokattam. Mid. East Res. Cent. Ain Shams Univ., Sci. Res. Ser. 5, 65–78.

Moustafa, A.R., Yehia, M.A., Abdel Tawab, S., 1985. Structural setting of the area east of Cairo, Maadi, and Helwan. Mid. East Res. Cent. Ain Shams Univ., Sci. Res. Ser. 5, 40–64.

Najid, D., Westphal, M., Hernandez, J., 1981. Paleomagnetism of Quaternary and Miocene lavas from North-East and Central Morocco. J. Geophys. 49, 149–152.

Patel, J.P., Raja, P.K.S., 1979. Paleomagnetic results from the Narosura and Magadi Volcanics of Kenya. Phys. Earth Planet. Inter. 19, P7–P14.

Pouchan, P., Roche, A., 1971. Etude paleomagnetique de formations volcaniques du Territoire des Afars et Issas. C. R. Acad. Sci. Paris Ser. D 272, 531–534.

Rasmussen, D.T., Bown, T.M., Simons, E.L., 1992. The Eocene–Oligocene transition in continental Africa. In: Prothero, D.R., Berggren, W.A. (Eds.), Eocene–Oligocene Climatic and Biotic Evolution. Princeton Univ. Press, pp. 548–566.

Reilly, T.A., Raja, P.K.S., Mussett, A.E., Brock, A., 1976. The palaeomagnetism of Late Cenozoic Volcanic rocks from Kenya and Tanzania. Geophys. J. R. Astr. Soc. 45, 483–494.

Said, R., 1962. Geology of Egypt. Amsterdam, Elsevier, 377 pp.

Scalera, G., Favali, P., Florindo, F., 1996. Palaeomagnetic database: the effect of quality filtering for geodynamic studies. In: Morris, A., Tarling, D.H. (Eds.), Palaeomagnetism and Tectonics of the Mediterranean Region. Geol. Soc. Spec. Public. No. 105, pp. 225–237.

Schult, A., 1974. Palaeomagnetism of Tertiary volcanic rocks from the Ethiopian Southern Plateau and the Danakil Block. J. Geophys. 40, 203–212.

Schult, A., Hussain, A.G., Soffel, H.C., 1981. Palaeomagnetism of Upper Cretaceous Volcanics and Nubian Sandstones of Wadi Natash, SE Egypt and implications for the Polar Wander Path for Africa in the Mesozoic. J. Geophys. 50, 16–22.

Schult, A., Soffel, H., 1973. Palaeomagnetism of Tertiary Basalts from Libya. Geophys. J. R. Astr. Soc. 30, 373–380.

Simons, E.L., 1990. Discovery of the oldest known anthropoidean skull from the Paleogene of Egypt. Science 247, 1567–1569.

Simons, E.L., Wood, A.E., 1968. Early Cenozoic faunas, Fayum Province, Egypt. Bull. Peabody Mus. Nat. Hist., Yale Univ. 28, 1–105.

Strougo, A., 1985. Eocene stratigraphy of the eastern Greater Cairo (Gebel Mokattam-Helwan) area. Mid. East Res. Cent. Ain Shams Univ., Sci. Res. Ser. 5, 1–39.

Tarling, D.H., Abdeldayem, A.L., 1996. Palaeomagnetic-pole er-

rors and a 'small-circle' assessment of the Gondwanan polar-wander path. Geophys. J. Int. 125, 115–122.

Tauxe, L., Besse, J., LaBrecque, J.L., 1983. Paleolatitudes from DSDP Leg 73 sediment cores: implications for the apparent polar wander path for Africa during the Late Mesozoic and Cenozoic. Geophys. J. R. Astr. Soc. 73, 315–324.

Van Couvering, J.A., Harris, J.A., 1991. Late Eocene age of the Fayum mammal faunas. J. Hum. Evol. 21, 241–260.

Van der Voo, R., 1990. Phanerozoic paleomagnetic poles from Europe and North America and comparisons with continental reconstructions. Revs. Geophys. 28, 167–206.

Walton, D., 1996. Magnetic overprints and their removal. Phys. Earth Planet. Inter. 94, 145–148.

Westphal, M., 1993. Did a large departure from the geocentric axial dipole hypothesis occur during the Eocene? Evidence from the magnetic polar wander path of Eurasia. Earth Planet. Sci. Lett. 117, 15–28.

Youssef, M., Cherif, O., Boukhary, M., Mohamed, A., 1984. Geological studies on the Sakkara area, Egypt. N. Jb. Geol. Paläont. Abh. 168, 125–144.

Zalat, A.A., 1987. Paleontologic and stratigraphic studies on the Eocene limestones at Gebel Mokattam and Mishgigah, Egypt. MSc thesis, Tanta Univ., 308 pp.

Zijderveld, J.D.A., 1967. A.C. demagnetization of rocks: analysis of results. In: Collinson, D.W. et al. (Eds.), Methods in Palaeomagnetism. Elsevier, Amsterdam, pp. 254-286.